

TASK 9 SUBMISSION

Competitor Demand Prediction Using Job Postings

Hiring Intelligence Report

NVIDIA: what its job postings reveal

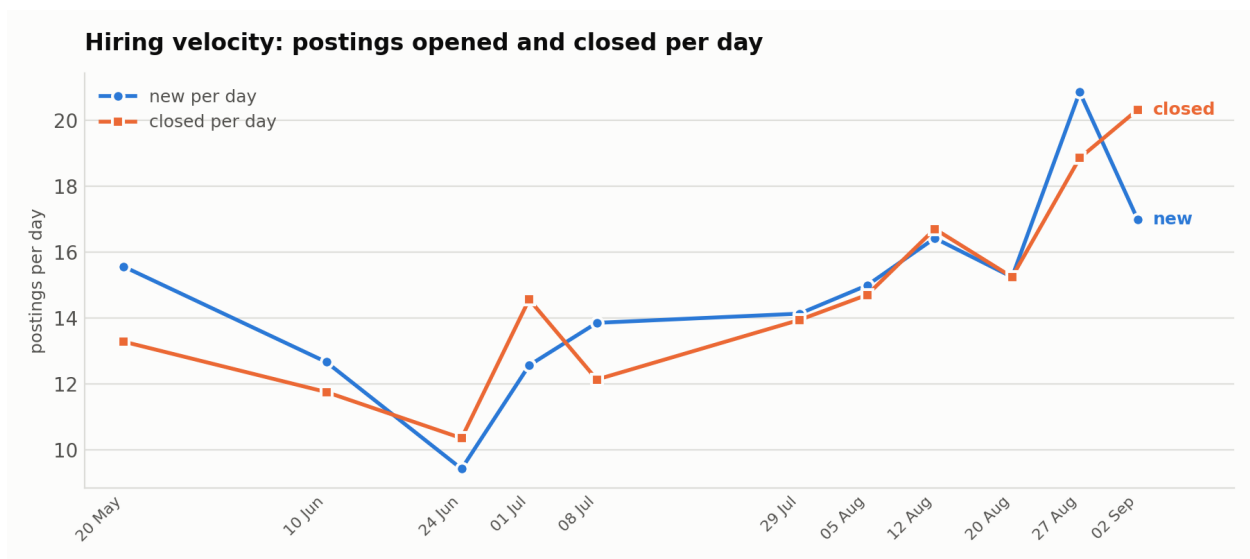
Prepared by	Nayab Khalid
Company	NVIDIA
Window	13 May to 2 September 2026
Evidence base	1,745 postings across 12 weekly observations
Date	20 September 2026
Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Executive summary

1. **NVIDIA is not expanding its open-role count; it is turning it over faster.** Roughly 416 roles stay open at any time, up 4.6% over four months, while both the arrival and closure rates rose about 25%.
2. **NVIDIA hires the people who build the machine, not the people who use it.** Hardware, systems and silicon account for 885 of 1,793 skill mentions. ML frameworks - PyTorch, TensorFlow, JAX - account for 14.
3. **The rotation is toward the field.** Product and go-to-market rose from 13.7% to 18.8% of skill mentions while research fell from 16.1% to 11.3%, and Solutions Architecture is now the largest single skill in the corpus at 11% of postings. That is a company scaling commercial deployment, not one expanding research.

1 Hiring position

Measure	Value	Source
Open roles	395 to 433, mean 416	Task 5
Net change over four months	+18 (+4.6%)	Task 5
New postings per day	mean 14.8, rising about 25%	Task 5
Closed per day	mean 14.7, rising in step	Task 5
Median time a posting stays open	14 days	Task 5
Forecast, 4 weeks ahead	416 (394 to 439)	Task 7



Postings opened and closed per day. The two lines never diverge by more than 3.3 per day.

New and closed track each other at **every single observation**. This is replacement hiring held at a steady level, not growth. A company adding headcount would show arrivals outrunning departures; these move in lockstep while both accelerate.

2 Skill demand

Skill	Share of postings	Skill	Share of postings
Solutions Architecture	11.0%	Performance Engineering	6.5%
Networking	8.1%	Deep Learning	4.6%
Cloud Infrastructure	7.6%	Silicon Verification	4.5%
Datacenter Systems	6.8%	SoC Design	3.5%

Rising fastest, comparing the second half of the window with the first:

Skill	First half	Second half	Skill	First half	Second half
Inference and Serving	3	14	HPC	14	24
Compilers	11	25	Physical Design	6	13
Agentic AI	17	31	AI Safety	0	7

AI Safety appearing from literal zero is the cleanest signal in the set: unlike a skill that merely grows, it cannot be a sampling artefact.

Concentrated at senior level: Storage Systems 94% senior or above, C++ 91%, Analog and Mixed Signal 90%, Signal and Power Integrity 90%. That is consistent with a scarce talent pool, although this data cannot prove it.

3 Competitive position

NVIDIA is unlike the other three companies in this study. Hardware plus silicon is 49.4% of NVIDIA's skill mentions against 8.1% for Google. Each company's own cloud appears in its own postings: GCP for Google, Azure for Microsoft, neither for NVIDIA.

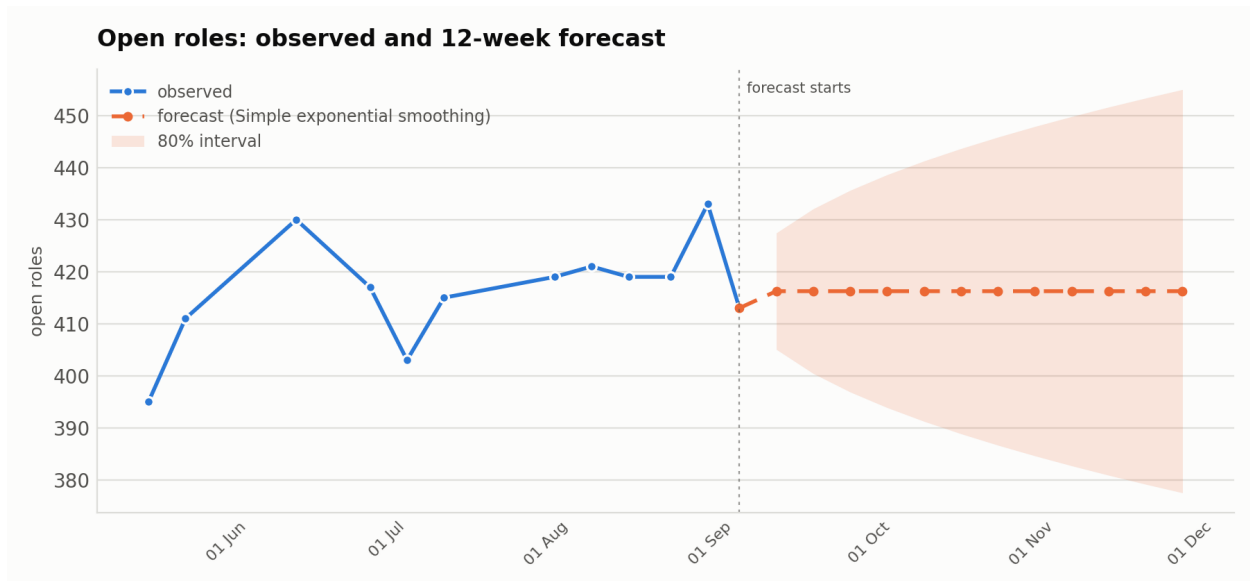
Two caveats that matter more than the numbers

First, the skill taxonomy was extended by me for NVIDIA's vocabulary, which inflates NVIDIA's apparent distinctiveness; on the pre-extension taxonomy NVIDIA and Google are level, at 21.2% and 22.3% coverage. Second, NVIDIA cannot appear in the only stable similarity matrix at all, because its source carries no description text. Both are documented in Tasks 6 and 8, and both are fixable by the team.

4 What a decision-maker should do with this

If you are	Then
Competing for silicon talent	Expect a standing pool of about 416 NVIDIA roles turning over in roughly two weeks. Analog, mixed-signal and signal-integrity roles are about 90% senior, so the contested segment is experienced hires, not graduates
Watching NVIDIA's strategy	Track the shift toward Solutions Architecture and Developer Relations. It suggests commercial deployment of AI infrastructure is scaling faster than research
Planning training	Inference and serving, compilers and agentic AI are rising fastest. All three sit between model research and hardware - the deployment layer
Benchmarking salary	Not possible from this data. The source carries no salary field for NVIDIA

5 Outlook



Open roles, observed and forecast, with the 80% interval.

The forecast is flat: about 416 open roles, with a four-week interval of 394 to 439. The twelve-week interval is 377 to 455, wide enough that it only says "about the same". Notably, the upward trend in arrivals measured in Task 5 **does not forecast**: in backtesting, the models that extrapolate a trend performed worst. The trend is real in the sample and is not projectable.

6 What this data cannot tell you

#	Limitation
1	Closed does not mean filled. Nothing separates a filled role from a cancelled or relisted one, so fast turnover is not evidence of fast hiring.
2	European bias. The source curates EU-office employers, so this is NVIDIA's European hiring rather than global.
3	Titles, not descriptions. About one skill per posting against 15 to 25 from a description. A skill absent here means the title did not say it, not that NVIDIA does not need it.
4	Four months. No seasonality can be estimated, and the 12-week forecast interval is too wide to plan against.
5	Third-party derived data. NVIDIA's own Terms of Service ruled out first-party collection, as established in Task 1.

7 The honest bottom line

The strongest claims in this report - flat stock with accelerating churn, hardware dominance, and the rotation toward the field - rest on NVIDIA's own data and survive every robustness check applied to them.

The cross-company claims do not yet. Task 6 showed that the four-way comparison currently measures my taxonomy as much as it measures the companies, and Task 8 showed that the similarity ranking flips when the taxonomy changes. Both are fixable, and the fix is the same in each: every member extends the taxonomy for their own company, then everyone re-runs. Reporting that plainly is more useful to a decision-maker than a confident number that would not survive contact with a second analyst.

Traceability

Every figure in this report traces to a committed artefact: Task 4 `nvidia_skill_features_20260902.csv`, Task 5 `nvidia_weekly_trend_20260902.csv`, Task 6 `basis_comparison.csv`, Task 7 `forecast_summary.csv`, Task 8 `matrix_comparison.csv`.

Declaration	No personal or sensitive data is present in any output. This report synthesises the artefacts committed in Tasks 4 to 8; no new data was collected.
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Date	20 September 2026